

Simulation Documentation

Periodic Fourier–AR(1) Spatio-Temporal Gaussian Process
Stats 669 Research Reference

Pranav Tikkawar

May 2026

Contents

1	Model Overview	2
2	Parameter Specification	2
2.1	Spatial Spectrum: Matérn-like Fourier Variances	2
2.2	Temporal Persistence: Power-Law Decay	3
2.3	Phase Rotation	3
2.4	Innovation Correlation	3
2.5	Parameter Summary Table	3
3	Temporal Dynamics: Block-Diagonal AR(1)	4
3.1	Transition Matrix for Mode $j \geq 1$	4
3.2	Mode $j = 0$: Scalar AR(1)	4
3.3	Innovation Covariance	4
4	Space–Time Covariance	4
5	Simulation Algorithm	5
6	Spatial Field Evaluation: Direct vs. FFT	6
6.1	Method 1: Direct Summation	6
6.2	Method 2: Inverse FFT on a Dense Regular Grid (Recommended)	6
6.3	Method 3: Adaptive Truncation	7
7	Stationary Properties	7
8	Connection to Continuous-Time SDE	8
9	Validation Checklist	8
10	Recommended Default Parameters	9
11	Open Questions	9

1 Model Overview

We simulate a real-valued Gaussian spatio-temporal process

$$\{X_t(x) : x \in [0, 2\pi], t \in \mathbb{Z}\}$$

defined on a one-dimensional *periodic* spatial domain (a circle) with discrete time. The process is represented as a truncated real-valued Fourier series,

$$X_t(x) = a_0(t) + \sum_{j=1}^J [a_j(t) \cos(jx) + b_j(t) \sin(jx)], \quad x \in [0, 2\pi], \quad (1)$$

where $J \in \mathbb{Z}_+$ is the Fourier truncation level and

$$\mathbf{Z}_t = (a_0(t), a_1(t), b_1(t), a_2(t), b_2(t), \dots, a_J(t), b_J(t))^T \in \mathbb{R}^{2J+1}$$

is the state vector of Fourier coefficients at time t .

Markov property. The state vector $\mathbf{Z}_t$ is Markov: the distribution of $\mathbf{Z}_{t+1}$ depends only on $\mathbf{Z}_t$. This Markov property holds for the *full* coefficient vector; individual spatial values $X_t(x_0)$ at a fixed location are *not* Markovian.

2 Parameter Specification

2.1 Spatial Spectrum: Matérn-like Fourier Variances

The marginal variance allocated to each Fourier mode j follows a Matérn-like spectral density evaluated at integer frequencies:

$$\tilde{\sigma}_j^2 = (\kappa^2 + j^2)^{-(\nu+1/2)}, \quad j = 0, 1, \dots, J, \quad (2)$$

where

- $\nu > 0$ is the Matérn *smoothness* parameter (larger $\nu \Rightarrow$ smoother fields),
- $\ell > 0$ is the spatial *range* parameter,
- $\kappa = \sqrt{2\nu}/\ell$ is the inverse-scale (larger $\kappa \Rightarrow$ shorter range).

These are normalized to sum to a target total variance σ_{tot}^2 :

$$\sigma_j^2 = \sigma_{\text{tot}}^2 \cdot \frac{\tilde{\sigma}_j^2}{\sum_{k=0}^J \tilde{\sigma}_k^2}. \quad (3)$$

The resulting $\{\sigma_j^2\}_{j=0}^J$ are the *stationary variances* of the cosine and sine coefficients for each mode.

2.2 Temporal Persistence: Power-Law Decay

Each mode j has its own AR(1) persistence coefficient $\rho_j \in (0, 1)$. High-frequency modes decay faster in time (non-separability). We use an *algebraic* (power-law) parameterization:

$$\rho_j = \frac{1}{(1 + \lambda j^\alpha)^\beta}, \quad j = 1, \dots, J, \quad (4)$$

with $\rho_0 \in (0, 1)$ set separately, and hyperparameters:

- $\lambda > 0$: overall temporal decay strength,
- $\alpha > 0$: frequency exponent (how strongly persistence changes with j),
- $\beta > 0$: decay exponent (slower polynomial decay for smaller β).

The continuous-time (SDE) interpretation is $\rho_j = e^{-\lambda_j}$ where $\lambda_j = \beta \log(1 + \lambda j^\alpha)$ is the mode-specific damping rate.

2.3 Phase Rotation

For each mode $j \geq 1$, a rotation angle $\theta_j \in \mathbb{R}$ controls temporal phase drift. The default parameterization is

$$\theta_j = \frac{\theta_0}{j}, \quad j = 1, \dots, J, \quad (5)$$

for scalar hyperparameter θ_0 . Setting $\theta_j = 0$ removes phase rotation entirely.

2.4 Innovation Correlation

Within each mode $j \geq 1$, the cosine and sine innovations are correlated with parameter $\phi_j \in (-1, 1)$. The default parameterization is

$$\phi_j = \frac{\phi_0}{1 + j}, \quad j = 1, \dots, J, \quad (6)$$

for scalar hyperparameter $\phi_0 \in (-1, 1)$. Setting $\phi_j = 0$ makes innovations independent within each mode.

2.5 Parameter Summary Table

Symbol	Name	Domain	Role
J	Fourier truncation	$\mathbb{Z}_+$	Spatial resolution
ν	Matérn smoothness	$(0, \infty)$	Spatial regularity
ℓ	Spatial range	$(0, \infty)$	Correlation length
σ_{tot}^2	Total variance	$(0, \infty)$	Field scale
ρ_0	Constant-mode persistence	$(0, 1)$	Long-run mean reversion
λ	Temporal decay strength	$(0, \infty)$	Overall temporal memory
α	Frequency exponent	$(0, \infty)$	Persistence-frequency slope
β	Power-law exponent	$(0, \infty)$	Decay curvature
θ_0	Rotation hyperparameter	$\mathbb{R}$	Phase drift rate
ϕ_0	Correlation hyperparameter	$(-1, 1)$	Cosine-sine coupling

3 Temporal Dynamics: Block-Diagonal AR(1)

3.1 Transition Matrix for Mode $j \geq 1$

The pair $(a_j(t), b_j(t))$ evolves as a bivariate AR(1):

$$\begin{pmatrix} a_j(t+1) \\ b_j(t+1) \end{pmatrix} = M_j \begin{pmatrix} a_j(t) \\ b_j(t) \end{pmatrix} + \begin{pmatrix} \eta_{a,j,t} \\ \eta_{b,j,t} \end{pmatrix}, \quad (7)$$

where the 2×2 transition (rotation-scaling) matrix is

$$M_j = \rho_j \begin{pmatrix} \cos \theta_j & -\sin \theta_j \\ \sin \theta_j & \cos \theta_j \end{pmatrix}. \quad (8)$$

Key properties of M_j :

- $M_j M_j^\top = \rho_j^2 I_2$ (orthogonal scaling — no cross-coupling in second moments),
- Eigenvalues: $\rho_j e^{\pm i\theta_j}$ (modulus $= \rho_j < 1$ ensures stationarity),
- $\theta_j = 0$: reduces to $\rho_j I_2$, cosine and sine evolve independently.

3.2 Mode $j = 0$: Scalar AR(1)

$$a_0(t+1) = \rho_0 a_0(t) + \eta_{0,t}, \quad \eta_{0,t} \sim \mathcal{N}(0, \sigma_0^2(1 - \rho_0^2)). \quad (9)$$

3.3 Innovation Covariance

The innovation covariance for mode $j \geq 1$ is determined by the *discrete-time Lyapunov equation*: stationarity $\Sigma_j = M_j \Sigma_j M_j^\top + \Sigma_{\varepsilon,j}$ with $M_j M_j^\top = \rho_j^2 I_2$ gives

$$\Sigma_{\varepsilon,j} = (1 - \rho_j^2) \Sigma_j.$$

Targeting stationary covariance $\Sigma_j = \sigma_j^2 \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix}$ yields:

$$\Sigma_{\varepsilon,j} = \sigma_j^2 (1 - \rho_j^2) \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix}. \quad (10)$$

Note: this result is *independent of* θ_j . The rotation angle affects temporal structure but not marginal variances or contemporaneous correlation.

4 Space–Time Covariance

For $\theta_j = 0$ (no rotation) and modes independent across frequencies, the space–time covariance is

$$C(h, k) = \mathbb{E}[X_t(x) X_{t+k}(x+h)] = \sum_{j=0}^J \sigma_j^2 \rho_j^{|k|} \cos(jh). \quad (11)$$

Because ρ_j depends on j , this cannot be factored as $C_{\text{space}}(h) \cdot C_{\text{time}}(k)$, so the model is *nonseparable*.

When $\theta_j \neq 0$, the lag- k transition matrix is

$$M_j^k = \rho_j^k \begin{pmatrix} \cos(k\theta_j) & -\sin(k\theta_j) \\ \sin(k\theta_j) & \cos(k\theta_j) \end{pmatrix},$$

so the temporal cross-covariance of (a_j, b_j) acquires oscillatory structure:

$$\text{Cov}(a_j(t), a_j(t+k)) = \sigma_j^2 \rho_j^k \cos(k\theta_j) + \sigma_j^2 \phi_j \rho_j^k \sin(k\theta_j).$$

5 Simulation Algorithm

Algorithm 1 Periodic Fourier–AR(1) Spatio-Temporal Simulation

Require: Parameters: $T, J, \nu, \ell, \sigma_{\text{tot}}^2, \rho_0, \lambda, \alpha, \beta, \theta_0, \phi_0$

Ensure: Coefficient arrays $\mathbf{a} \in \mathbb{R}^{T \times (J+1)}$, $\mathbf{b} \in \mathbb{R}^{T \times (J+1)}$

- 1: **Precompute parameters:**
 - 2: $\sigma_j^2 \leftarrow$ equation (3) for $j = 0, \dots, J$
 - 3: $\rho_j \leftarrow$ equation (4) for $j = 1, \dots, J$; ρ_0 set directly
 - 4: $\theta_j \leftarrow \theta_0/j$ for $j = 1, \dots, J$
 - 5: $\phi_j \leftarrow \phi_0/(1+j)$ for $j = 1, \dots, J$
 - 6: $M_j \leftarrow$ equation (8) for each $j \geq 1$
 - 7: $\Sigma_{\varepsilon,j} \leftarrow$ equation (10) for each $j \geq 1$
 - 8: **Initialize at $t = 1$ from stationary distribution:**
 - 9: $a_0(1) \leftarrow \mathcal{N}(0, \sigma_0^2)$
 - 10: **for** $j = 1, \dots, J$ **do**
 - 11: $(a_j(1), b_j(1))^\top \leftarrow \mathcal{N}_2\left(\mathbf{0}, \sigma_j^2 \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix}\right)$
 - 12: **end for**
 - 13: **Forward simulation for $t = 2, \dots, T$:**
 - 14: **for** $t = 2, \dots, T$ **do**
 - 15: $a_0(t) \leftarrow \rho_0 a_0(t-1) + \mathcal{N}(0, \sigma_0^2(1 - \rho_0^2))$
 - 16: **for** $j = 1, \dots, J$ **do**
 - 17: $\boldsymbol{\eta} \sim \mathcal{N}_2(\mathbf{0}, \Sigma_{\varepsilon,j})$
 - 18: $(a_j(t), b_j(t))^\top \leftarrow M_j(a_j(t-1), b_j(t-1))^\top + \boldsymbol{\eta}$
 - 19: **end for**
 - 20: **end for**
 - 21: **Evaluate spatial field** (choose one of two methods below)
-

6 Spatial Field Evaluation: Direct vs. FFT

6.1 Method 1: Direct Summation

Given coefficient arrays $\mathbf{a}, \mathbf{b}$ and a spatial grid $\{x_n\}_{n=1}^N$, evaluate directly:

$$X_t(x_n) = a_0(t) + \sum_{j=1}^J [a_j(t) \cos(jx_n) + b_j(t) \sin(jx_n)]. \quad (12)$$

Cost: $O(NJ)$ per time step. Suitable for small N or irregular grids.

6.2 Method 2: Inverse FFT on a Dense Regular Grid (Recommended)

The professor recommends evaluating on a *dense regular grid* and using the Inverse FFT (IFFT) for speed. The key identity is:

$$X_t\left(\frac{2\pi k}{N}\right) = \text{Re} \left[\sum_{j=0}^{N-1} \tilde{c}_j e^{2\pi i j k / N} \right] = N \cdot \text{IFFT}(\tilde{\mathbf{c}})[k],$$

where the complex coefficient vector $\tilde{\mathbf{c}} \in \mathbb{C}^N$ is packed as:

$$\tilde{c}_j = \begin{cases} a_0(t) & j = 0, \\ \frac{a_j(t) - i b_j(t)}{2} & j = 1, \dots, J, \\ \frac{a_{N-j}(t) + i b_{N-j}(t)}{2} & j = N - J, \dots, N - 1 \quad (\text{conjugate symmetric}), \\ 0 & \text{otherwise.} \end{cases} \quad (13)$$

Step-by-step procedure.

1. Choose grid size N (power of 2; require $N \geq 4J$ to avoid spatial aliasing).
2. Pack $\tilde{\mathbf{c}}$ from (a_j, b_j) using equation (13).
3. Compute $\mathbf{X}_t = \text{Re}[N \cdot \text{IFFT}(\tilde{\mathbf{c}})]$ in $O(N \log N)$.
4. To evaluate at arbitrary points $x^* \notin \{2\pi k/N\}$, interpolate from the dense grid.

Nyquist constraint. Frequencies $j > N/2$ cannot be represented on a grid of size N . Since we zero-pad all $j > J$ components in $\tilde{\mathbf{c}}$, choosing $N \geq 4J$ guarantees that:

- no aliasing occurs (frequencies $j \leq J \leq N/4 < N/2$),
- interpolation from the dense grid is accurate.

Computational comparison.

Method	Cost per time step	Example ($J=50, N=512$)
Direct summation	$O(NJ)$	$512 \times 50 = 25,600$ ops
IFFT	$O(N \log N)$	$512 \times 9 = 4,608$ ops
Speedup	—	$\approx 5.6\times$

For $N = 2048, J = 200$: speedup $\approx 18\times$.

6.3 Method 3: Adaptive Truncation

Choose J adaptively so the truncation error is below tolerance ε :

$$\sum_{j=J+1}^{\infty} \tilde{\sigma}_j^2 / \sum_{j=0}^{\infty} \tilde{\sigma}_j^2 < \varepsilon.$$

For the Matérn spectrum with $\nu > 0$, the tail sum converges since $\tilde{\sigma}_j^2 = O(j^{-2\nu-1})$ as $j \rightarrow \infty$. A safe rule of thumb: set $J \approx 5\kappa = 5\sqrt{2\nu}/\ell$.

7 Stationary Properties

At stationarity, the following hold for all t and all $j = 1, \dots, J$:

$$\text{Var}(a_j(t)) = \sigma_j^2, \tag{14}$$

$$\text{Var}(b_j(t)) = \sigma_j^2, \tag{15}$$

$$\text{Cov}(a_j(t), b_j(t)) = \sigma_j^2 \phi_j, \tag{16}$$

$$\text{Corr}(a_j(t), a_j(t+k)) = \rho_j^{|k|}. \tag{17}$$

These are the **validation targets** for simulation diagnostics. Note: equation (17) gives the ACF of the *coefficients* $a_j(t)$, not the amplitude $R_j(t) = \sqrt{a_j(t)^2 + b_j(t)^2}$. The amplitude ACF differs from $\rho_j^{|k|}$ in general.

Marginal field variance. The total marginal variance of the field at any point x is

$$\text{Var}(X_t(x)) = \sigma_0^2 + \sum_{j=1}^J \sigma_j^2 (1 + \phi_j \cdot 2 \cos(jx) \cdot \sin(jx)).$$

If $\phi_j = 0$ for all j , this simplifies to $\sum_{j=0}^J \sigma_j^2 = \sigma_{\text{tot}}^2$, independent of x .

8 Connection to Continuous-Time SDE

The discrete-time AR(1) is the *exact discretization* (Euler–Maruyama at $\Delta t = 1$) of the continuous-time Ornstein–Uhlenbeck system:

$$d \begin{pmatrix} a_j(t) \\ b_j(t) \end{pmatrix} = \underbrace{\begin{pmatrix} -\lambda_j & -\theta_j \\ \theta_j & -\lambda_j \end{pmatrix}}_{F_j} \begin{pmatrix} a_j(t) \\ b_j(t) \end{pmatrix} dt + \sqrt{2\lambda_j} \sigma_j d \begin{pmatrix} W_{a,j}(t) \\ W_{b,j}(t) \end{pmatrix}, \quad (18)$$

where $\lambda_j = -\log(\rho_j) > 0$ is the damping rate, and $W_{a,j}, W_{b,j}$ are independent standard Wiener processes.

The matrix exponential $e^{F_j \Delta t}$ with $\Delta t = 1$ recovers M_j exactly:

$$e^{F_j} = e^{-\lambda_j} \begin{pmatrix} \cos \theta_j & -\sin \theta_j \\ \sin \theta_j & \cos \theta_j \end{pmatrix} = \rho_j \begin{pmatrix} \cos \theta_j & -\sin \theta_j \\ \sin \theta_j & \cos \theta_j \end{pmatrix} = M_j.$$

The full infinite-dimensional SDE (stacking all modes) is:

$$d\mathbf{Z}(t) = \mathbf{F} \mathbf{Z}(t) dt + \mathbf{L} d\mathbf{W}(t),$$

where $\mathbf{F} = \text{diag}(-\lambda_0, F_1, F_2, \dots)$ is a block-diagonal operator and $\mathbf{L} = \text{diag}(\sqrt{2\lambda_0}\sigma_0, \sqrt{2\lambda_1}\sigma_1 I_2, \dots)$.

Connection to Solin & Särkkä (2014). The purely periodic case ($\lambda_j = 0$, $\theta_j = j\omega_0$) corresponds to Solin & Särkkä’s stochastic resonator model. Our model extends this by adding damping ($\lambda_j > 0$), which corresponds to their *quasi-periodic* kernel obtained by multiplying a periodic covariance by a Matérn covariance in time.

9 Validation Checklist

Run the following checks after simulation to confirm correctness:

- V1. Variance:** For each j , check $\text{Var}(a_j(t)) \approx \sigma_j^2$ and $\text{Var}(b_j(t)) \approx \sigma_j^2$. (Equation (14)–(15))
- V2. Correlation:** For each j , check $\text{Cor}(a_j(t), b_j(t)) \approx \phi_j$. (Equation (16))
- V3. ACF:** For each j , the empirical ACF of $a_j(t)$ should decay as $\rho_j^{|k|}$. (Equation (17))
- V4. Phase drift:** For each j , the unwrapped phase $\psi_j(t) = \text{atan2}(b_j(t), a_j(t))$ should drift linearly with slope $\approx \theta_j$ rad/step.
- V5. Spatial covariance:** The empirical spatial covariance $\hat{C}(h, 0)$ should match $\sum_j \sigma_j^2 \cos(jh)$.
- V6. Total variance:** $\sum_j \hat{\sigma}_j^2 \approx \sigma_{\text{tot}}^2$.
- V7. Normality:** Each $a_j(t)$ and $b_j(t)$ should pass a normality test (Q-Q plot or Shapiro–Wilk).
- V8. FFT accuracy:** For any time t , the IFFT-reconstructed field should match the direct-summation field to machine precision ($< 10^{-12}$).

10 Recommended Default Parameters

Parameter	Default	Notes
J	50	Increase to 100–200 for rough fields ($\nu < 1$)
ν	1.5	Twice mean-square differentiable
ℓ	1.0	Moderate spatial range on $[0, 2\pi]$
σ_{tot}^2	1.0	Unit variance
ρ_0	0.95	Long memory in the mean
λ	0.01	Slow global temporal decay
α	1.2	Slight superlinear frequency effect
β	1.0	Linear power-law
θ_0	0.3	Moderate phase drift at $j = 1$
ϕ_0	0.5	Moderate cosine–sine coupling
T	200	Sufficient for ACF estimation up to lag 50
N	512	FFT grid; $N = 4J \times 2 = 400$, round to 2^9

11 Open Questions

- Q1. Periodized Matérn spectrum:** Should σ_j^2 be computed by (a) sampling the continuous Matérn spectral density at integers j , or (b) computing the Fourier coefficients of the periodized Matérn covariance via Poisson summation? Option (b) is theoretically correct but more expensive.
- Q2. Amplitude ACF:** The theoretical ACF of the amplitude $R_j(t) = \sqrt{a_j^2 + b_j^2}$ is not simply $\rho_j^{|k|}$ (since R_j follows a Rice/Rayleigh distribution). What is the correct theoretical ACF of R_j ?
- Q3. Phase drift identifiability:** Is θ_j identifiable from data, or does it trade off with ϕ_j ?
- Q4. Spatial covariance ridge:** Does the empirical U-shaped spatial covariance at large lags correspond to Stein (2005)’s ridge artifact for nonseparable models?
- Q5. Parameterization for inference:** For Whittle likelihood estimation, should $\{\rho_j, \theta_j, \phi_j\}$ be free per-mode, or constrained via smooth functions of j ?